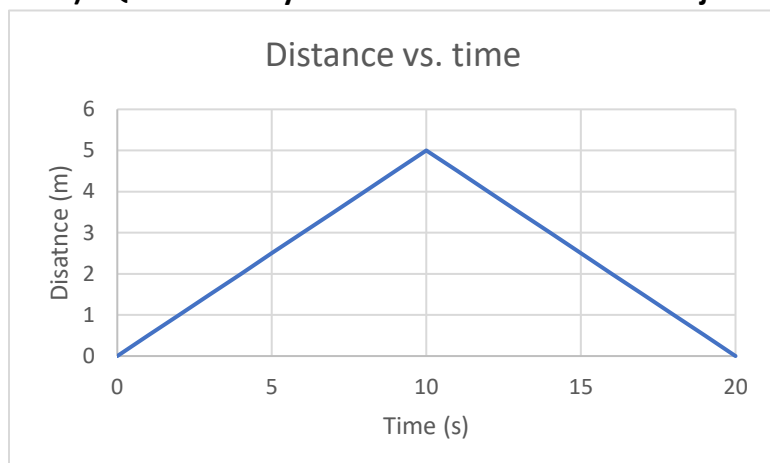


Graphs of Constant Speed Motion: Matching Graphs to Motion Scenario

Synopsis: Motion graphs *quantitatively* encapsulate the motion of an object – we can learn the position and speed of the object as a function of time. In this Note, you are tasked to 1) Describe the motion in words from a motion graph (distance vs. time or speed vs. time, 2) Sketch a motion graph from a description of an object's motion.

Performance Expectations: Give a quantitative description of an object's motion from a motion graph. Draw the motion graph from a description of an object's motion.

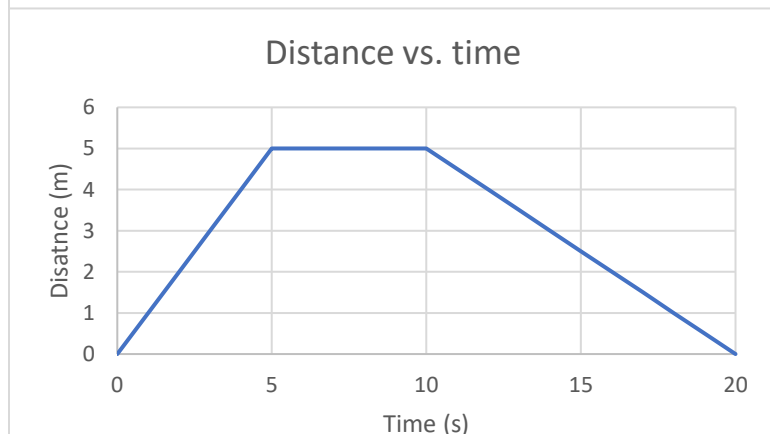
1) Quantitatively describe the motions of an object in the graphs (Must answer questions)



How far? Forward or backward? How fast?

- 0 – 10 s:

- 10 – 20 s:

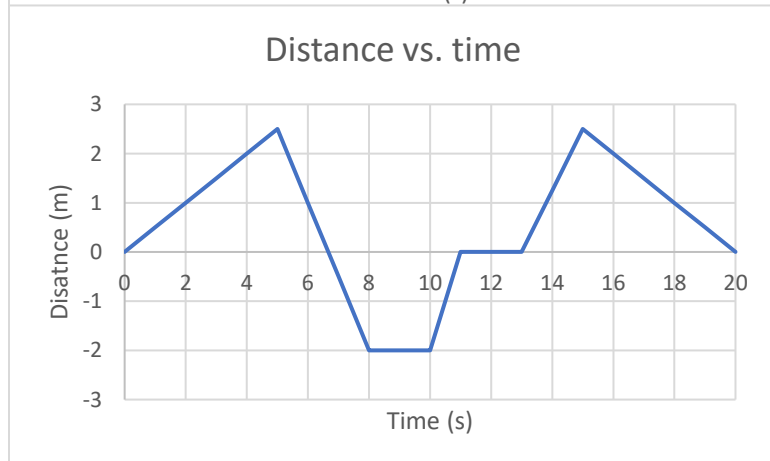


How far? Forward or backward? How fast?

- 0 – 5 s:

- 5 – 10 s:

- 10 – 20 s:



How far? Forward or backward? How fast?

- 0 – 5 s:

- 5 – 8 s:

- 8 – 10 s:

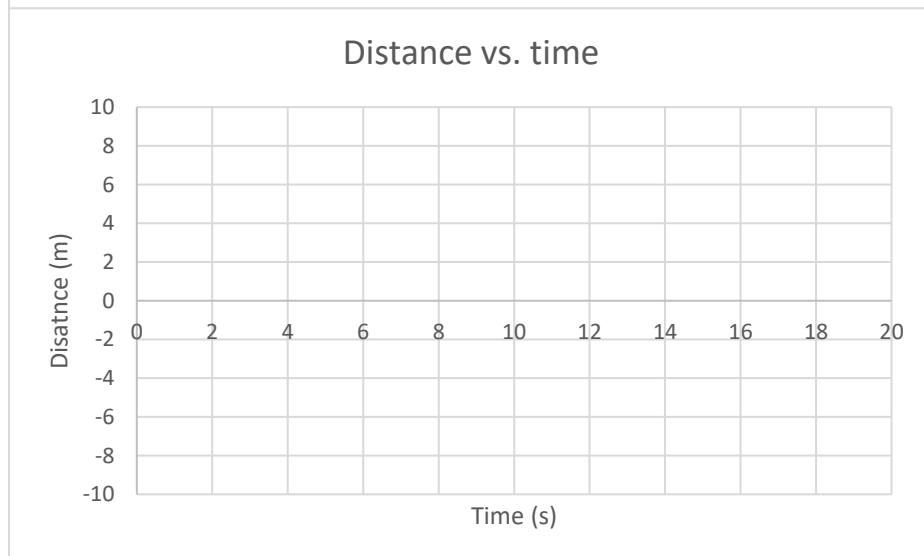
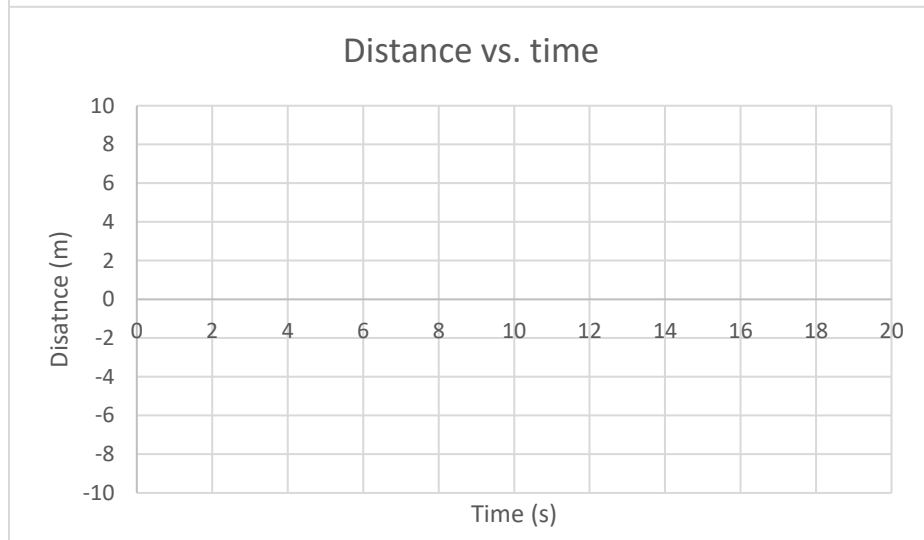
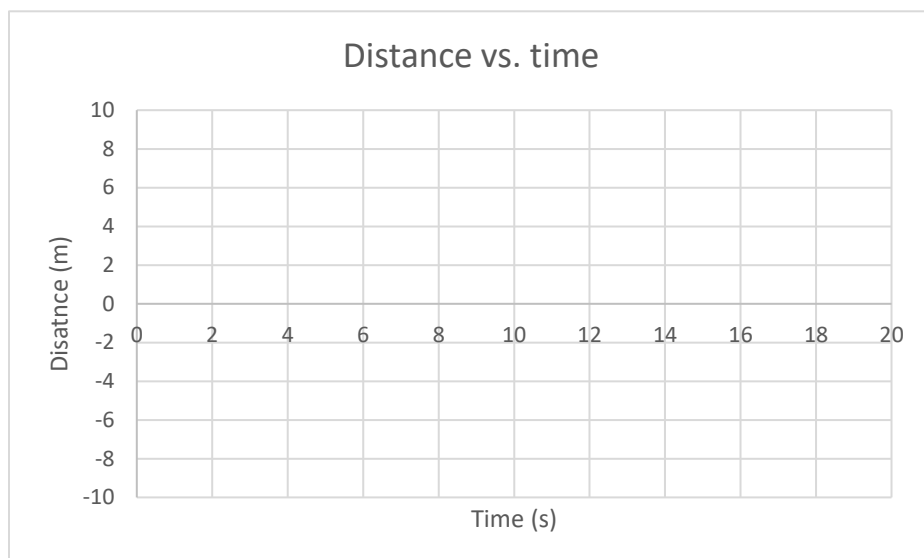
- 10 – 11 s:

- 11 – 13 s:

- 13 – 15 s:

- 15 – 20 s:

2) Draw the graph matching the description of the motion.



- Move forward at 1 m/s for 2s.
- Stop for 2s.
- Move forward at 2 m/s for 2s.
- Stop for 4 s.
- Move forward at 0.5 m/s 4s.
- Move backward at 1 m/s for 6 s.

Q) Where am I?

- Move backward at 1 m/s for 4s.
- Stop for 4 s.
- Move forward at 2 m/s for 4s.
- Stop for 2 s.
- Move backward 4 m in 6 s.

Q) Where am I?

Q) What is my final speed?

- Move forward at 2 m/s for 4s.
- Stop for 2 s.
- Move backward at 2 m/s for 8 s.
- Move forward at 1 m/s for 6 s.

Q) Where am I?

Q) What is my final speed?

3) Draw corresponding speed vs. time and acceleration vs. time graphs.

